Generative Artificial Intelligence Enabled STEM Learning: Community Garden Practices and Insights

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***Abstract*—This study investigates the integration of generative artificial intelligence (GenAI) in STEM education within community garden ecosystems. Using case studies and questionnaire surveys, it analyzes how GenAI facilitates interdisciplinary and STEM learning mechanisms. Two key practice cases illustrate its efficacy in constructing interdisciplinary contexts and advancing STEM engagement. Findings indicate that GenAI enhances students' STEM interest, knowledge application, and innovative capabilities, although challenges such as technical instability remain. Future developments will prioritize intelligent and personalized pathways to drive educational transformation.**

***Index Terms*—generative artificial intelligence; STEM learning; community gardens; smart education; interdisciplinary integration; project-based learning**

I. Introduction

In the context of deep integration between technological revolution and industrial transformation, the rapid advancement of artificial intelligence (AI) is reshaping the global education landscape. As outlined in the White Paper on China's Smart Education [1], AI is driving systemic transformations in education through restructuring knowledge delivery models and research innovation paradigms. Since the Chinese Ministry of Education launched the National Education Digitalization Strategy in 2022, it has operated under the "3C" philosophy—connection first, content priority, and cooperation-driven—and focused on the "3I" strategic framework: integration, intelligence, and internationalization. Guided by this framework, the initiative has established a smart education ecosystem spanning all educational stages and scenarios, offering Chinese solutions for global education digitization.

The traditional education model has long suffered from structural bottlenecks in one-way knowledge transmission, limiting the cultivation of students' core competencies such as interdisciplinary problem-solving and innovative thinking. STEM learning, rooted in the integration of Science, Technology, Engineering, and Mathematics, emphasizes project-based practice to develop complex problem-solving skills [2], aligning closely with the White Paper's vision of "building a lifelong and inclusive education system." As a pivotal technology for educational intelligence, generative AI enables the creation of multimodal content (e.g., text, images, data models) that synergizes with STEM's interdisciplinary demands. It not only constructs immersive learning environments via virtual labs and intelligent data analysis but also delivers personalized resources based on students' cognitive profiles, advancing the ideal of "large-scale adaptive education [3]." This aligns with the White Paper's strategic mandate to "explore intelligent education and implement AI-empowered initiatives."

This study provides innovative value for the integration of generative artificial intelligence (GAI) and STEM education from both practical and theoretical dimensions.

II. Purpose and Methodology

Guided by the White Paper's objective of "strengthening the deep integration of AI and education," this study focuses on the application of generative AI in STEM education within community gardens, using the Xinhua Community Garden in Chunhua Street, Jiangning District, Nanjing, as an empirical case. This approach aligns with the White Paper's initiative to "promote digital education applications through real-world scenarios." By examining specific cases—such as "sensor data-driven planting maintenance" and "large-model image analysis for garden management optimization"—the research explores the integration and innovation of AI technologies with localized educational contexts, providing micro-level practical

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insights for implementing the Education Digitization Strategy Action 2.0.

This study employs a mixed-methods approach integrating multiple research strategies. The case study methodology involves selecting a relatively mature community garden as the research site, implementing a generative AI–based STEM learning program within existing garden activities, and conducting in-depth analyses of the GenAI system's application processes and real-world impacts. The questionnaire survey collects pre- and post-program data from students to quantify the effects of generative AI on learning outcomes through statistical analysis. Semi-structured interviews with students, teachers, and community garden administrators capture multi-stakeholder perspectives, providing qualitative insights to triangulate quantitative findings.

III. Case Studies of Generative Artificial Intelligence Applications in Community Gardens

Community Gardens is a site for community residents to carry out horticultural activities in a common and shared way, which is a carrier for enhancing community public participation, building harmonious community relations, bringing people and nature closer together, realizing organic renewal, and thus promoting community building and community governance [4], and also a good occasion for education.

A. Sensor data and AI-based planting maintenance recommendation system

1) System architecture and workflow: The sensor data- and AI-driven planting maintenance recommendation system, as depicted in Fig. 1, comprises four core components: the sensor layer, data transmission layer, cloud server layer, and user interaction layer. The sensor layer deploys multi-type environmental sensors, including soil moisture sensors, light intensity sensors, temperature sensors, and carbon dioxide concentration sensors. These sensors enable real-time monitoring of community garden plants' growing conditions, collecting data on soil moisture, light intensity, temperature, and CO_2 levels. Additionally, embedded image sensors capture plant imagery for deep analysis of growth metrics (e.g., plant height, leaf count) through computer vision algorithms.

The data transmission layer employs IoT protocols to encrypt sensor-collected data and transmit it via wireless networks to the cloud server layer. Upon data reception, the cloud server layer performs cleaning and preprocessing to filter out noisy and anomalous data points. The validated data is then fed into a large language model (LLM), where machine learning algorithms conduct deep analytics to uncover underlying patterns and trends. By integrating plant growth knowledge bases with analytical insights, the system generates personalized planting recommendations tailored to specific species and environmental conditions.

The user interaction layer offers a unified interface for students, teachers, and community garden managers, supporting login via mobile apps, web portals, etc., to access real-time data and AI-generated maintenance recommendations. Users can submit feedback—including maintenance logs, queries, and suggestions—enabling the system to refine analytical models and recommendations iteratively. This establishes a closed-loop intelligent maintenance system, integrating data collection, transmission, analysis, recommendation generation, and feedback-driven optimization.

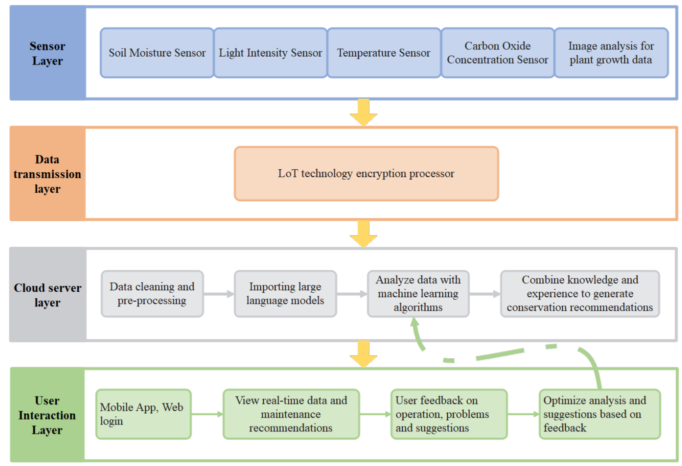

Fig. 1: Architecture of a sensor and AI-based system for generating plant care recommendations for community gardens.

2) Integration of STEM practices: In STEM learning practices, students engage with all system components. During sensor installation and calibration, they grasp sensor working principles and installation techniques, integrating knowledge from physics and electronics. In the data collection phase, students learn sensor functionalities, application scenarios, and data collection protocols, applying mathematical concepts to record and organize raw data. For data transmission and analysis, they explore IoT technologies and analytical algorithms, understanding cross-device data flow mechanisms and the logical foundations of algorithmic information extraction. When applying and validating maintenance recommendations, students analyze results, translate them into practical planting strategies, observe plant growth dynamics, and use scientific theories to interpret correlations between maintenance measures and plant development. Through these hands-on activities, students synthesize STEM knowledge and enhance their capability to apply multidisciplinary insights to real-world problem-solving. By addressing tasks through independent inquiry and collaborative teamwork, they develop project management, communication, and innovative thinking skills.

B. Large model-based analysis system

1) System functionality and operating logic: The large-model AI image analysis–based plant analysis system and user activity heatmap generation system serve three primary functions: plant information recognition, growth and planting recommendations, and user activity tracking. The system employs high-definition cameras installed in the community garden to collect real-time imagery of plants and visitor activities.

For the plant information recognition function, the system deploys a Vue3-based front-end interface for intelligent plant information query, integrated with a multimodal large language model (MLLM). Users obtain intelligent plant analysis by inputting query prompts and uploading plant images. The interface, structured with Ant Design Vue components, includes a prompt input field, image upload zone, query button, and results display area. Users can enter natural-language queries and upload PNG/JPEG images in the input field. Upon clicking "Start Analysis," the system encodes images into Base64 format, encapsulates prompts and images into a multimodal input structure, and invokes the open-source glm-4v-flash MLLM API for processing. The returned plant information—including species identification, growth characteristics, and care guidelines—is parsed and rendered in Markdown format for clear visualization.

For the growth planting suggestion function, analogous to the plant information recognition module, the system deploys a Vue3-based front-end interface integrated with a multimodal large language model (MLLM) for intelligent plant information query. The primary distinction is that users upload Excel files (.xlsx/.xls) containing plant growth environment data collected by smart sensors. After entering prompts in the input field and uploading the Excel file, clicking the "Start Analysis" button triggers the following process: the system parses the Excel data into JSON format via the XLSX library, encapsulates prompts and JSON data into a multimodal input structure, and transmits them to the glm-4v-flash MLLM API for processing. The resulting plant growth recommendations are parsed in Markdown format and displayed on the interface.

For the user activity tracking function, the system employs a multi-target tracking and visualization system based on the MMTracking framework to process videos for person tracking and headcount analysis. The core principle involves frame-by-frame reading of surveillance videos, using the ByteTrack algorithm for multi-target tracking on individual frames, and leveraging OpenCV and Matplotlib to visualize detection results. This includes drawing target bounding boxes, IDs, and confidence scores, as well as labeling the total and current number of people on each frame. Visualized frames are then merged into a tracking video with detection overlays via MMCV. Additionally, the system generates line graphs to depict temporal trends in total and current visitor numbers, enabling visualization of activity distribution and participation scale in the community garden. By integrating plant analysis results with user activity data, the system provides data-driven decision support for community garden managers and educators.

2) Application and value in learning activities: Regarding community garden management, the plant analysis results from the system enable managers to promptly grasp plant growth status and formulate reasonable maintenance plans. The user activity heatmap provides a foundation for managers to optimize garden layout and design activity areas. For instance, additional rest seats, ornamental facilities, etc., can be set up based on user activity hotspots to enhance community garden utilization efficiency and user experience.

From an educational perspective, the system holds significant application value. By examining plant image analysis results, students acquire knowledge about plant morphological structures and growth processes while cultivating observational skills and scientific thinking. Through learning about artificial intelligence, they understand the principles and applications of technologies like image recognition and data processing, thereby enhancing their information technology literacy. In the context of data analysis and problem-solving, students analyze user activity heatmaps to identify issues and propose solutions—such as investigating reasons for low activity in specific areas and optimizing garden layouts to attract more users—thereby developing data analysis and problem-solving competencies. The system can stimulate students' interest in science and technology, opening doors for them to explore the technological world.

IV. Specific Instructional Strategies for Generative Artificial Intelligence for STEM Learning

A. Building learning contexts for interdisciplinary knowledge integration

In community gardens, generative AI serves as a tool to integrate multidisciplinary knowledge and construct rich learning contexts for students. The core learning scenario of plant planting and maintenance leverages botany knowledge to elaborate on the growth and development processes, physiological characteristics of plants, etc., while incorporating environmental science to deeply analyze how environmental factors like soil, climate, and light affect plant growth. This interdisciplinary approach helps students establish connections between different knowledge systems and enhances their comprehensive understanding of the complex interactions in plant growth environments.

B. Project-driven learning activity-based design

The community garden program's objectives are firmly focused on fostering students' STEM literacy. By engaging in the construction and management of community gardens, students can deepen their understanding of the scientific principles underlying plant growth, master the application scenarios of technologies such as sensors and the Internet of Things, and utilize engineering thinking to participate in garden design and optimization. Throughout this process, their capabilities in mathematical analysis and data processing are enhanced.

The project tasks encompass multiple dimensions, featuring rich layering and diversity, as depicted in Figure 2.

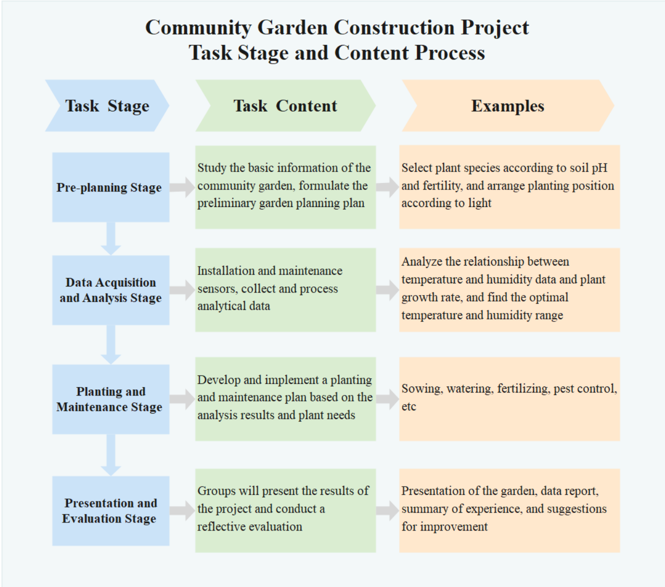

Fig. 2: Community Garden Construction Project Task Phase and Content Process.

In Figure3, the project implementation is divided into multiple phases, each with distinct tasks and objectives. These phases are closely interconnected, facilitating the gradual progression and completion of the project.

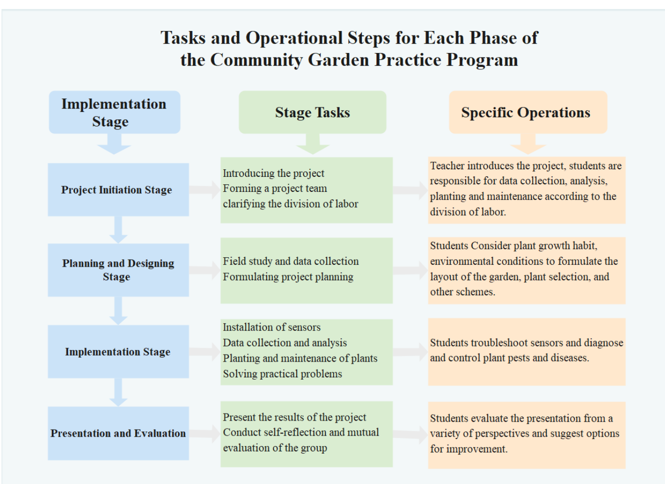

Fig. 3: Tasks and Operational Steps for Each Stage of the Community Garden Practice Program.

Through the design of such project-driven learning activities, students can engage in scientific research and engineering practice processes, foster their interest in and passion for STEM subjects, and establish a solid foundation for their future academic and career development.

V. Assessment and Reflection

A. Indicators and methods for assessing learning outcomes

To comprehensively and scientifically evaluate the impact of generative AI on students' learning outcomes in STEM education within community gardens, this study establishes a series of evaluation indicators and adopts a mixed-methods approach for assessment in Figure4.

In terms of knowledge acquisition, students are evaluated on their comprehension and retention of knowledge in science, technology, engineering, and mathematics (STEM) disciplines. For instance, their understanding of plant growth principles, sensor mechanisms, and data analysis methods is assessed through specially designed test questions. In a plant growth knowledge test, questions such as "Briefly describe the photosynthesis process and its significance for plant growth" and "List three common sensors and explain their applications in community gardens" were posed. Students' responses to these questions were utilized to gauge their mastery level of the subject matter.

In terms of attitudinal change, the assessment primarily focuses on evaluating students' shifts in interest and attitudes toward STEM subjects.

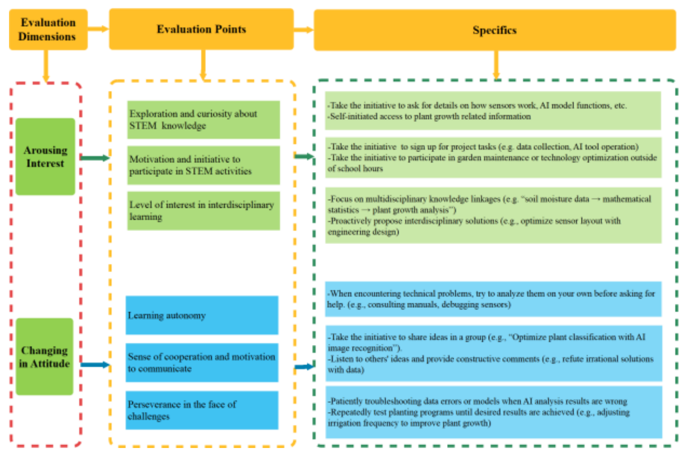

Fig. 4: Evaluation Framework for Interest and Learning Attitude in Community Garden STEM Practices.

In terms of assessment methods, multiple approaches including tests, work evaluation, and questionnaires were employed in Figure5.

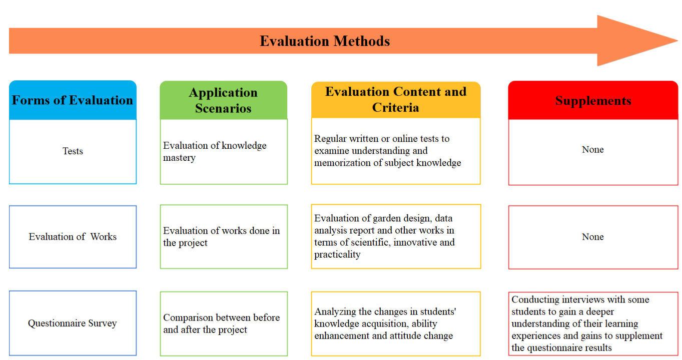

Fig. 5: Multi-dimensional Evaluation Methods and Criteria for STEM Programs.

B. Problems and directions for improvement

Despite the remarkable achievements of generative AI in STEM education within community gardens, its practical application has unveiled critical challenges that demand in-depth analysis and targeted solutions.

In terms of technical reliability, generative AI still has substantial room for improvement. The analytical outputs of generative AI are not entirely accurate, often yielding recommendations that deviate from real-world conditions. Regarding students' over-reliance on generative AI, this issue warrants significant attention. To guide students to use generative AI appropriately, teachers should enhance educational guidance to foster students' independent learning awareness and critical thinking skills. During the teaching process, educators can pose open-ended questions to encourage students to think independently before leveraging generative AI tools for assistance. Additionally, students should be instructed to analyze and evaluate the answers and suggestions from generative AI, learn to discern the authenticity and merits of information, and thereby enhance their information literacy [5].

VI. Conclusions

Guided by the "Artificial Intelligence Enabled Education" concept outlined in the White Paper on China's Smart Education, this study investigates the application of generative AI in STEM education within community gardens. Through two practical projects—"Sensor Data-Driven Planting and Maintenance System" and "Big Model Image Analysis for Garden Management"—in the context of community gardens, the research comprehensively demonstrates the pivotal role of AI technology in fostering interdisciplinary learning and project-based learning (PBL) [6]. These initiatives exemplify how generative AI can facilitate knowledge integration across science, technology, engineering, and mathematics while addressing real-world challenges in localized educational settings [7].

This study serves as both a micro-level implementation of the White Paper on China's Smart Education strategy and an empirical reference for the localized innovation of intelligent education. By embedding generative AI in community garden practices, it illustrates how educational digitization can be operationalized through tangible [8], context-specific applications, aligning with the national "3C" (connection, content, cooperation) and "3I" (integration, intelligence, internationalization) frameworks.

References

[1] "White Paper on China's Smart Education (2025)". (2025, May 17). World Digital Education Conference. https://wdec.smartedu.cn/publication?id=1.

[2] R. W. Bybee, The Case for STEM Education: Challenges and Opportunities. Arlington, VA: NSTA Press, 2013.

[3] N. Yu and Y. Zhang, "The influence of ChatGPT/AIGC on education: New frontiers of great power games," J. East China Normal Univ. (Educ. Sci.), vol. 41, no. 7, pp. 15–25, 2023.

[4] X. Ding, Z. Zhao, J. Zheng, X. Yue, H. Jin, and Y. Zhang, "Community gardens in China: Spatial distribution, patterns, perceived benefits and barriers," Sustain. Cities Soc., vol. 85, p. 103991, 2022.

[5] Z. Yang, J. Wang, D. Wu, and X. Chen, "Exploring the impact of ChatGPT/AIGC on education and strategies for response," J. East China Normal Univ. (Educ. Sci.), vol. 41, no. 7, pp. 26–35, 2023.

[6] Y. Li, "Reconstructing Science PBL Teaching Mode Empowered by Generative Artificial Intelligence: From the Integrated Perspective of Resources, Process and Evaluation", *Primary and Secondary School IT Education*, vol. 2025, no. 4, pp. 41 - 44, 2025.

[7] D. Wu, Z. Guo Yuan, X. Nan Wang, "Practical Research on Artificial Intelligence Cross - Empowering Secondary School Science Education", *Innovative Talent Education*, vol. 2025, no. 2, pp. 27 - 34, 2025.

[8] W. Hong Wu, Y. Yi Ye, B. Xu, "Innovation and Practice of Digital Teaching Reform Based on the Integration of Science and Education and Blended Teaching", *Industry and Science Forum*, vol. 24, no. 13, pp. 130 - 132, 2025.